IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP PRESENTATION

Artificial Intelligence & Machine Learning

STUDENT

Roushan Kant Singh

ROLL NUMBER

25SCS1003002793

PROGRAMME

B.Tech CSE | Batch 2025-29

ORGANIZATION

Codec Technologies Pvt. Ltd.

ROLE

Artificial Intelligence Intern

DURATION

15 Jun 2026 – 15 Jul 2026

INTRODUCTION

Internship Overview

A one-month Artificial Intelligence internship undertaken with Codec Technologies Pvt. Ltd., conducted in association with the National Internship Portal (Ministry of Education) and Google for Education, providing structured, hands-on exposure to the applied AI/ML pipeline.

PROGRAM FOCUS AREAS

- Artificial Intelligence
- Machine Learning
- Deep Learning
- NLP
- Reinforcement Learning
- Capstone Projects

ROLE

Artificial Intelligence Intern

DURATION

15/06/2026 – 15/07/2026 (1 Month)

MODE

Pan India, Hybrid

Practical competencies were built across the end-to-end data lifecycle — exploratory analysis, time-series forecasting, CNNs for vision, NLP, Q-learning, and applied capstone engineering.

HOST ORGANIZATION

About Codec Technologies Pvt. Ltd.

A specialized technology software and industrial training enterprise approved by AICTE and ICAC, delivering professional upskilling and applied industrial training across emerging computing paradigms.

AICTE VERIFICATION ID

CORPORATE6759d549ce59e1733940553

NCS IDENTIFICATION

E19E86-0116588288923

QUALITY CERTIFICATION

ISO 9001:2015 Certified

PROGRAM LEADERSHIP

Dr. Anurag Shrivastava (Program Manager / Talent Acquisition Manager)

INSTITUTIONAL PARTNERSHIPS

- **Google for Education Partner**
- **AICTE & ICAC Approved**
- **National Internship Portal (Ministry of Education)**
- **ISO 9001:2015 Certified Organization**

LEARNING GOALS

Internship Objectives

Python for AI Development

Programming workflows using NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn

Data Preprocessing & Visualization

Cleaning, transforming, and visualizing multi-variate datasets

Machine Learning Foundations

Supervised learning, model evaluation and optimization

Deep Learning & LSTM Forecasting

Stacked LSTM networks for sequential financial time-series

Natural Language Processing

Text processing and sentiment analysis using NLTK & spaCy

Reinforcement Learning

Q-learning based sequential decision-making

Feature Engineering

Sliding windows, scaling, imputation, SMOTE class balancing

Capstone Implementation

End-to-end applied projects with quantitative model evaluation

TECHNOLOGY STACK

Technologies & Tools Used

PROGRAMMING

- Python 3.10+

DATA PROCESSING

- NumPy
- Pandas

VISUALIZATION

- Matplotlib
- Seaborn

MACHINE LEARNING

- Scikit-Learn
- XGBoost
- Imbalanced-Learn (SMOTE)

DEEP LEARNING

- TensorFlow
- Keras

NLP

- NLTK
- spaCy

DEV ENVIRONMENT

- Google Colab
- Jupyter Notebook

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TOOLS & LIBRARIES ACROSS THE PIPELINE

METHODOLOGY

Weekly Curriculum Timeline

WEEK 1

Foundations of AI & Python

- AI concepts
- Python for AI
- Data preprocessing & visualization

WEEK 2

Machine Learning Basics

- Supervised learning
- Model evaluation & optimization

WEEK 3

Deep Learning & CNN

- Neural networks
- CNN fundamentals
- Image recognition
- Basic CNN model

WEEK 4

Advanced AI Applications

- NLP, text processing, sentiment analysis (NLTK, spaCy)
- Reinforcement Learning, Q-learning
- Model evaluation metrics

Capstone Projects

Stock Price Prediction + Weather / Rainfall Prediction

Internship Certification

PIPELINE DESIGN

System Architecture

MULTI-MODAL DATA INGESTION

- Stock Financial Time-Series
- Meteorological Sensor Data

PREPROCESSING & FEATURE PIPELINE

- Missing-value imputation & normalization
- Encoding, scaling & sliding-window tensors

AI MODELING CORE

- Deep LSTM Neural Network
- Ensemble Classifiers (RF & XGBoost)

QUANTITATIVE EVALUATION & INFERENCE

- RMSE, MAE, R^2, Directional Accuracy
- Accuracy, Precision, Recall, F1, ROC-AUC

CAPSTONE PROJECT 1

Stock Price Trend Prediction using Deep LSTM

DOMAIN

FinTech · Deep Learning · Continuous Time-Series Forecasting

PROBLEM

Equity markets exhibit non-linearity, volatility, and long-range temporal dependence that traditional econometric models (ARIMA/GARCH) struggle to capture.

OBJECTIVE

Predict next-day closing price (Pt+1) from a historical N-day lookback window using stacked LSTM networks.

DATASET & SPLIT

Open, High, Low, Close, Adjusted Close, Volume · **80% Train / 20% Test (chronological)**

1. Historical Stock Data
2. Preprocessing (MinMaxScaler)
3. 60-Day Lookback Window
4. Stacked LSTM Network
5. Price Prediction
6. **Evaluation (RMSE, MAE, R^2)**

LSTM Architecture & Results

LSTM Layer — 50 Units *return_sequences = True*

Dropout — Rate 0.20

LSTM Layer — 50 Units *return_sequences = True*

Dropout — Rate 0.20

LSTM Layer — 50 Units *return_sequences = False*

Dropout — Rate 0.20

Dense Output — 1 Unit *Linear activation*

TRAINING CONFIGURATION

50 Epochs · Batch Size 32 · Adam Optimizer · MSE Loss · Early Stopping

TEST-SET PERFORMANCE (OUT-OF-SAMPLE)

2.14
RMSE

1.62
MAE

0.962
R^2 SCORE

87.4%
DIRECTIONAL ACCURACY

The model explains 96.2% of price variance and correctly forecasts next-day direction in 87.4% of test cases.

CAPSTONE PROJECT 2

Weather Data Analysis & Rainfall Prediction

DOMAIN

Environmental Data Science · Applied Meteorology · Supervised Classification

DATASET

140,000+ historical daily weather observations across 23 atmospheric parameters

GOAL

Predict binary precipitation occurrence — Rain Tomorrow: Yes / No

KEY ATMOSPHERIC FEATURES

Temperature

Humidity

Pressure

Wind Speed

Sunshine

Cloud Cover

PREPROCESSING PIPELINE

- Missing-value imputation (median / modal)
- Outlier capping via IQR method
- Cyclic date & pressure/temperature feature engineering
- One-Hot Encoding of categorical features
- SMOTE class-imbalance correction
- StandardScaler feature scaling

140,000+

DAILY WEATHER OBSERVATIONS USED

Model Benchmark & Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	82.6%	71.3%	52.8%	0.606	0.842
Random Forest (200 Trees)	86.1%	78.4%	61.2%	0.687	0.891
Tuned XGBoost Classifier	**88.4%**	**81.2%**	**67.5%**	**0.737**	**0.924**

Best Performing Model: Tuned XGBoost Classifier

88.4% Accuracy · 81.2% Precision · ROC-AUC 0.924

Key predictive drivers: Humidity (3 PM), Atmospheric Pressure (3 PM), Cloud Cover & Sunshine Duration.

Overall Performance & Achievements

MAJOR CAPSTONE PROJECTS

Stock Price Prediction (Deep LSTM)

0.962
R^2 SCORE

87.4%
DIRECTIONAL ACCURACY

Weather Rainfall Prediction (XGBoost)

88.4%
ACCURACY

0.924
ROC-AUC

ADDITIONAL CURRICULUM EVALUATION METRICS

(module exercises within the training curriculum, distinct from the two major capstone projects above)

Computer Vision

91.8%
TOP-1 ACCURACY

NLP Sentiment Analysis

89.5%
POLARITY F1-SCORE

Reinforcement Learning

~ Epoch 350
POLICY CONVERGENCE

VERIFICATION

Weekly Task Completion — Portal Screenshots

Verified screenshots from the Codec Technologies learning portal, confirming progressive completion of all curriculum modules and capstone work.

Week 1 — Foundations of AI & Python

Week 2 — Machine Learning Basics

Week 3 — Neural Networks & CNNs

Week 4 (Part 1) — NLP & Sentiment Analysis

Week 4 (Part 2) — Reinforcement Learning

Capstone Work & Certificate Claim

OFFICIAL RECORDS

Internship Certificate & Offer Letter

CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

CERTIFICATE OF INTERNSHIP

This certificate is awarded to

ROUSHAN KANT SINGH

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC Approved internship program as

Artificial Intelligence Intern

Conducted from 15/06/2026 to 15/07/2026

Google for Education Partner

Dr. Anurag Shrivastava

Program Manager

Internship Completion Certificate

ROLE	ORGANIZATION	DURATION
AI Intern	Codec Technologies Pvt. Ltd.	15/06/2026 – 15/07/2026

CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

INTERNSHIP OFFER LETTER

Dear ROUSHAN KANTSINGH,

We are pleased to offer you a 1 Month internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 15/06/2026 to 15/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.
Best regards,

Sincerely,
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 15/06/2026

Dr. Anurag Shrivastava

Google for Education Partner

ISO 9001 CERTIFIED

www.codectechnologies.in support@codectechnologies.in Chandivali IT hub, Mumbai

Internship Offer Letter

CONCLUSION

Key Takeaways from the Internship

- Practical, end-to-end understanding of AI/ML workflows
- Applied ML classification & benchmarking using XGBoost
- Hands-on experience in data preprocessing & model development
- Exposure to CNNs, NLP and Reinforcement Learning
- Deep Learning proficiency through stacked LSTM forecasting
- Quantitative model evaluation using industry-standard metrics

THANK YOU

Questions & Discussion

Roushan Kant Singh

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